

Literature Review

What were the three main neuron classes discussed? What type of signals do they send and what functions do these neurons serve in small neuronal circuits?

Excitatory neurons:

- Send excitatory signals
- Transmit information forward through the circuit

Inhibitory neurons:

- Send inhibitory signals
- Feedforward inhibition, Feedback inhibition, lateral inhibition, mutual inhibition

Modulatory neurons:

- Use neuromodulators
- Regulate circuit state and behavior

Neurons are discrete cells connected to each other through specific points of contact, rather than forming a continuous tissue like muscle or epithelium. What functional advantages does this discrete, network-based architecture offer the nervous system?

- Specific connectivity, flexible information processing, modularity, Plasticity, efficient wiring

What happens when the balance between excitatory and inhibitory cells is off?

- Network becomes hyperactive or neural activity becomes suppressed

What structural motifs do you expect to spontaneously emerge in a random neuronal culture? Which ones would not occur?

Likely:

- Feedforward chain, simple recurrent loops, mutual inhibition, local excitation-inhibition motifs

Not likely:

- Topographic maps
- Discrete parallel processing channels
- Biased input–segregated output neuromodulatory systems
- Large-scale hierarchical circuit architectures